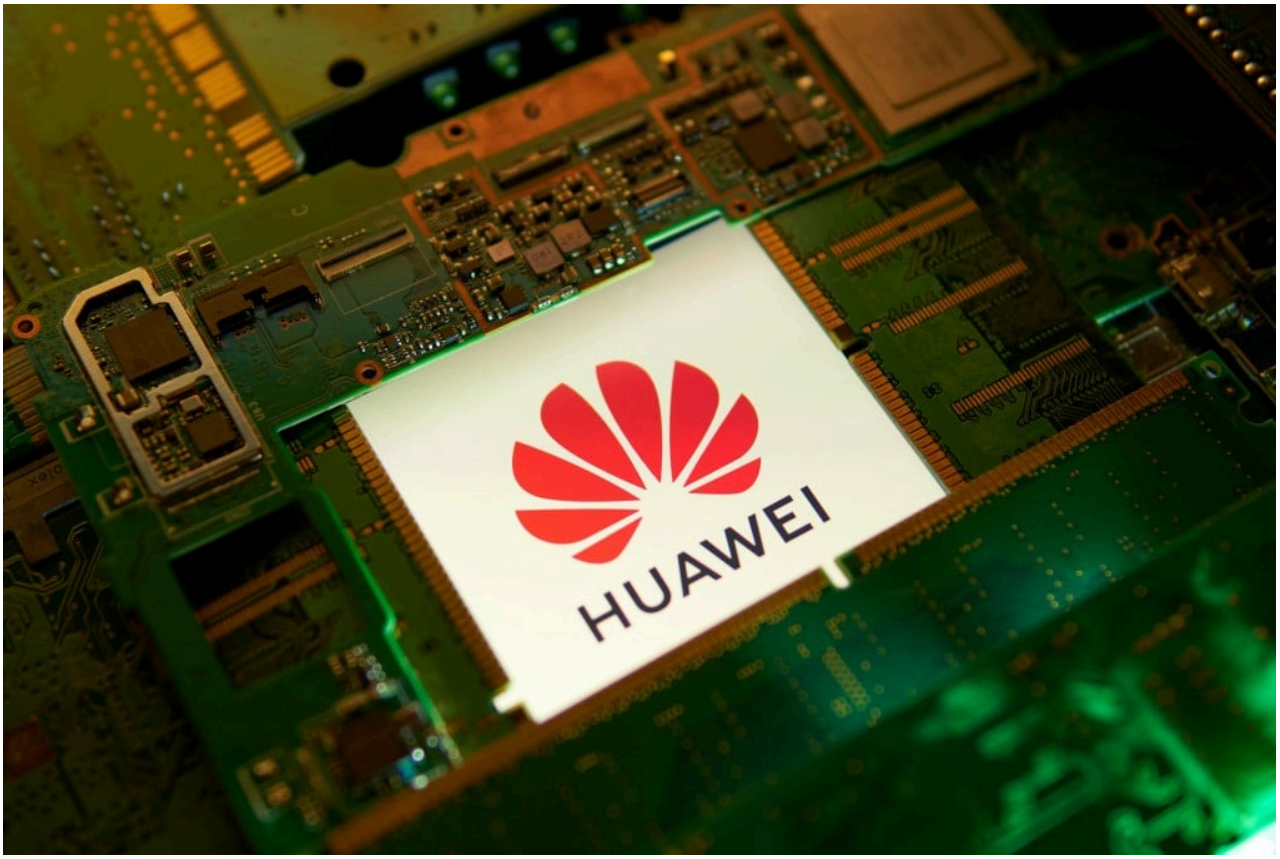
Tech war: Huawei's computing cluster to power AI projects, challenge US platforms

scmp.com/tech/tech-war/article/3326432/tech-war-huaweis-computing-cluster-power-ai-projects-challenge-us-platforms

Ann Cao September 23, 2025



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Published: 9:00am, 23 Sep 2025

Huawei Technologies' quest for greater computing power for artificial intelligence projects is expected to rest on the firm's latest supernode computing cluster systems, as these take on rival platforms from Nvidia and xAI.

The Atlas 950 and Atlas 960 SuperPoDs, as well as the Atlas 950 and Atlas 960 SuperClusters, are expected to help the Shenzhen-based firm "circumvent the limitations in China's chip manufacturing process", Huawei deputy chairman Eric Xu Zhijun, who serves as the privately held company's rotating chairman from April to September, told Chinese media on the sidelines of the Huawei Connect 2025 conference in Shanghai last week.

By the fourth quarter of next year, Huawei plans to release the Atlas 950 SuperPoD that is powered by 8,192 Ascend neural processing unit (NPU) cards – optimised for AI neural networks and machine learning tasks and applications.

The Atlas 960 SuperPoD, powered by up to 15,488 Ascend 960 NPU cards, will be available by the end of 2027, according to Huawei.

The Atlas 950 and 960 SuperClusters are large-scale computing clusters that comprise multiple Huawei SuperPoDs. The Atlas 950 SuperCluster, available by the end of 2026, will have more than 500,000 Ascend NPUs, while the Atlas 960 SuperCluster will have more than 1 million Ascend NPUs.

With the “world’s most powerful” SuperPoDs and SuperClusters, Huawei has what it takes to provide “abundant computing power for ongoing, rapid advancements in AI, both now and in the future,” Xu said.

Huawei deputy chairman Eric Xu Zhijun speaks at the opening of the Huawei Connect 2025 conference last Thursday. Photo: Handout

Huawei’s latest computing systems reflect the company’s increased confidence in designing advanced chips to power AI development projects, following years of holding its semiconductor development plans close to its chest since the firm was added to Washington’s trade blacklist in 2019.

These new systems extend the supernode concept, which puts “Huawei at the top of the list in terms of card capacity among publicly available products”, said Charlie Zheng, chief economist at Chinese company Samoyed Cloud Technology Group.

The products are also expected to rival the capabilities of Nvidia’s NVL platform and the Colossus supercomputer developed by Elon Musk’s xAI.

Xu said the Atlas 950 SuperPoD would remain the most powerful of its kind in the world for “at least many years to come”.

That system is 56.8 times the scale, 6.7 times the total computing power, 15 times the memory capacity and 62 times the interconnect bandwidth compared to Nvidia’s NVL144, which will be available in the second half of next year, and the NVL576, scheduled for release in 2027, according to Xu.

Compared with xAI’s Colossus, the world’s largest and most powerful supercomputer designed for AI training, the Atlas 950 SuperCluster is 2.5 times larger in scale and 1.3 times more powerful, Xu said.

Underpinning Huawei’s latest computing systems is UnifiedBus, an interconnect protocol for SuperPoD. At the conference last week, Huawei also released the technical specifications for UnifiedBus 2.0 to push wider adoption of the protocol among industry partners.

The UnifiedBus 2.0 is expected to contribute to the development of a domestic ecosystem for Huawei’s supernode and cluster computing technologies, according to a research note from Huaxi Securities.

Such advancement would also benefit domestic semiconductor design firms such as Cambricon Technologies and contract chipmakers led by Semiconductor Manufacturing International Corp, the Huaxi note said.

The latest products came as Huawei has been making progress in pushing the industrial adoption of its CloudMatrix 384 system, which was launched earlier this year and is based on the Supernode384 architecture.

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